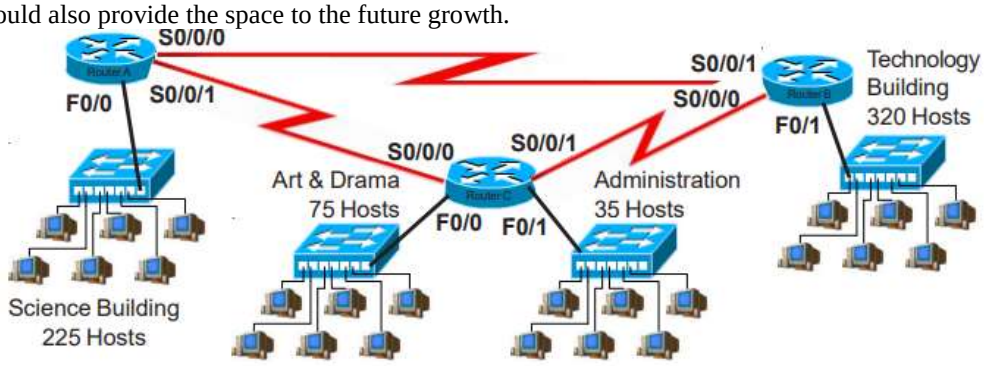
School of Electronics Engineering
Winter Semester 2022-23
Continuous Assessment Test - II

Programme : B.Tech (ECE)
Course Code : ECE4008 Course Name : COMPUTER COMMUNICATION
Faculty Name : P PRAKASAM
Slot : C1 + TC1 Class ID : VL2022230500952
Date of Exam : 14.03.2023

Max 50 Marks

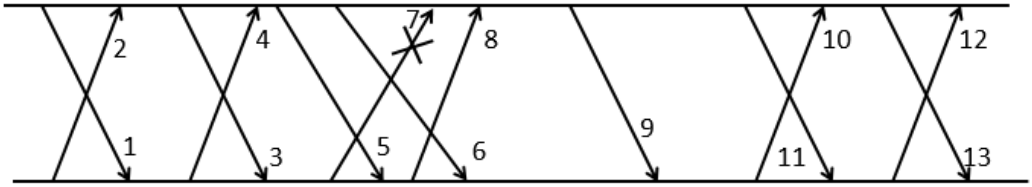
Q.No	Question	Max Marks	CO	BL
1	A government organization has been assigned with IP address 135.70.0.0. It requires minimum of four thousands usable hosts per subnet. Find the following to design the subnet. a) Default subnet mask 255.255.0.0 b) Custom subnet mask 255.255.240.0 c) Total number of subnets 14 d) Total number of host addresses 4096 e) Number of usable addresses 4094 f) Number of bits borrowed 4 g) Subnet range for 12th subnet 135.70.192.1 - 135.70.207.254 h) Subnet ID for 5th subnet 135.70.80.0 i) First valid host address for 14th subnet 135.70.224.1 j) Broadcast address for 10th subnet 135.70.175.255	10	CO 5	BL3
2	A University is equipped with four different buildings namely, Administration Building, Technology Building, Science Building, Art & Drama Building. Using an IP address 120.0.0.0, configure the network such that it will provide the link to required number of hosts per subnet. Also it has been projected that there is 40% growth in all blocks in future. Your configuration should also provide the space to the future growth.  Determine the 1. Address class A 2. Custom subnet mask 255.240.0.0 3. Minimum number of subnets needed 7 4. Extra subnets required for 40% growth 3 5. Total number of subnets needed 10 6. Minimum number of hosts per subnet needed 320 7. Extra hosts required for 40% growth 128 8. Total number of hosts per subnet required 448 9. IP range - Technology Building 10. IP range - Science Building	15	CO 5	BL4

11. IP range - Arts & Drama Building
12. IP range – Administration Building
13. IP - range between routers

120.16.0.1 - 120.31.255.254, 120.32.0.1 - 120.47.255.254, 120.48.0.1 - 120.63.255.254, 120.64.0.1 - 120.79.255.254, 120.80.0.1 - 120.95.255.254, 120.96.0.1 - 120.111.255.254, 120.112.0.1 - 120.127.255.254, 120.128.0.1 - 120.143.255.254, 120.144.0.1 - 120.159.255.254, 120.160.0.1 - 120.175.255.254, 120.176.0.1 - 120.191.255.254, 120.192.0.1 - 120.207.255.254, 120.208.0.1 - 120.223.255.254, 120.224.0.1 - 120.239.255.254

3 — 10 — CO 3 — BL4

The following figure corresponds to an HDLC ABM frame exchange. Consider 3 bits to represent the sequence number. Complete the diagram by completing the labeling of the frame exchanges



1) BI00 2) AI00 3) xIxx 4)xIxx 5) xIxx 6) xIxx 7) xIxx 8) xIxx 9) xSREJx 10) xIxx 11) xIxx 12) xIxx 13) xIxx

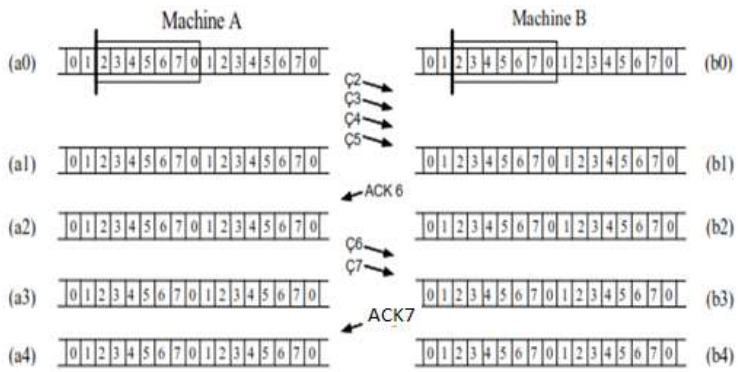
Solution

1) BI00 2) AI00 3) BI11 4)AI11 5) BI22 6) BI32 7) AI22 8) AI33 9) ASREJ2 10) AI24 11) BI42 12) AI45 13) BI54

4 — 6+4 — CO 3 — BL3

i). Go-back-N sliding window protocol is used between the data link layers of machines A and B with a window size of 7 frames at both sides. The figure below shows the a0 and b0 states of the sliding windows of machines A and B, respectively, just before Frame 2 is sent by A. The same figure also shows the sequence numbers of the frames sent by A and acknowledgements sent by B.

Indicate the frame numbers that should be contained in the sliding windows and buffer memories at the sender and receiver for the states a1, b1, a2, b2, a3, b3, a4, b4.



Machine A			Machine B		
State	Place of the Buffer Line	Frames in the Sliding Window	State	Place of the Buffer Line	Frames in the Sliding Window
a1	Between 1 and 2	6,7,0	b1	Between 1 and 2	6.7.0
a2	Between 5-6	6,7,0,1,2,3,4	b2	Between 5-6	6,7,0,1,2,3,4
a3	Between 5-6	0, 1, 2, 3,4	b3	Between 5-6	0, 1, 2, 3,4
a4	Between 6-7	7, 0, 1, 2, 3,4,5	b4	Between 6-7	7, 0, 1, 2, 3,4,5

ii). From the received data '8E 7D 8D AD AE 5E 20 E5 8D AE 8D AE AD 8D AD 7E 5E 8E' destuff using the flag byte '8E' and control byte as '8D'

7D 8D AE 5E 20 E5 8E 8E AD 8D 7E 5E

5 — 5 — CO 3 — BL3

In the data link layer of a computer network, bit-oriented framing and bit stuffing method is applied. Assuming that 011110 bit string is used as the starting and ending delimiter.

a). At the sender, if the bit string delivered by the network layer to the data link layer is given as

	follows, obtain the bit string at the data field of the sender's data link layer. 1 0 1 1 1 1 1 1 0 0 1 1 1 0 0 0 1 1 1 1 011110 1 0 1 1 1 0 1 1 1 0 0 0 1 1 1 0 0 0 0 1 1 1 0 1 011110 b). At the receiver, if the bit string delivered by layer 1 to layer 2 is as follows, obtain the bit string before bit stuffing. ~~0 1 1 11 0~~ 0 0 1 0 1 0 0 1 11~~0~~ 0 11 1 ~~0~~ 11 1 ~~0~~ 0 0 1 11 ~~0~~ 0 1 11 ~~0~~ 1 0 0 ~~0 1 1 11 0~~			